

Ghost in the Machine: Exploring Designed Fear in HRI

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Abstract

This paper examines how robots can safely and intentionally evoke fear in humans. Drawing on popular culture, immersive theater, horror games, and theme parks, we highlight design strategies such as unpredictability, proximity violations, and narrative buildup. We discuss how robot morphology, motion, and ambient cues shape perceived threat, with implications for entertainment-focused Human-Robot Interaction (HRI) and for preventing unintentional fear in everyday interactions. The paper aims to spark discussion on leveraging fear as a meaningful, ethical, and safe emotional modality in human-robot interaction.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; Interaction design; • **Computer systems organization** → Robotics.

Keywords

Human-Robot Interaction; Social Robots; Recreational Fear

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1 Introduction

Robots are increasingly present in human environments, yet their design largely prioritizes safety, predictability, and comfort. While this approach minimizes negative emotional responses, it limits exploration of more complex interactions, including fear and suspense.

This paper argues that studying fear in Human-Robot Interaction (HRI) is valuable not only for designing robots that intentionally evoke fear in entertainment contexts such as horror media or theme parks, but also for understanding and preventing unintentional fear in everyday human-robot interactions. This position paper surveys relevant literature across psychology, media studies, and HRI to argue for the systematic study of fear as a designed interaction modality in human-robot interaction. We position fear as a meaningful and informative emotional response rather than a design failure, highlighting its potential both as a deliberate tool for entertainment and as a lens for understanding unintended emotional responses in everyday human-robot interactions. By clarifying these dual applications, the paper encourages discussion within the HRI community on how fear can be ethically, safely, and effectively incorporated into robot design.

2 Conceptual Foundations for Fear in HRI

2.1 Recreation fear

Recreational fear refers to the voluntary engagement with frightening experiences—most commonly through horror media (film, television, crime podcasts), thrill rides, and haunted attractions—for enjoyment or personal challenge [46]. Seemingly paradoxical, these practices align with longstanding theories of play behavior across species, which allow animals to rehearse threat-related skills in low-risk environments, such as when young pigs engage in mock fights [53]. Humans similarly use fear-based play to probe the boundaries of their emotional and physiological responses to stress, from playing hide-and-seek in childhood to consuming horror media as adults [6].

Haunted attractions offer a useful setting for studying these processes. They are purpose-built environments that use themed sets, atmospheric effects, and scare actors or automated scares to elicit fear, anxiety, disgust, and disquiet. This provides a controlled context in which researchers can observe how individuals regulate high-arousal states in real time [16]. Work in this area has shown that recreational fear engages mechanisms related to broader psychological models, such as clarifying how intolerance of



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uncertainty, the tendency to find ambiguous situations distressing, shapes approach behavior under controlled threat [23].

Empirical studies consistently demonstrate that moderate recreational fear in a safe environment can be pleasurable. Research identifies an inverted-U relationship between fear and enjoyment, showing that when arousal reaches an ideal level, participants report the greatest enjoyment, mirrored by optimal deviations in heart-rate dynamics [6]. Excessive or prolonged arousal, by contrast, is experienced as unpleasant. Additional findings suggest physiological and emotional benefits such as improved immune markers [13], reduced neural activity following high-arousal negative experiences, and increased relaxation after intense haunted-house encounters [30]. Horror media consumption has also been linked to greater physiological resilience during the COVID-19 pandemic, possibly due to repeated practice managing fear and uncertainty [47].

Despite this growing literature, robots intentionally designed to evoke fear remain largely unexplored. Robots possess physical presence, autonomous motion, and real-time responsiveness, enabling forms of high-arousal interaction not achievable with static props or virtual media. Leveraging these capabilities for controlled fear induction represents a novel opportunity for HRI and entertainment-oriented design research.

2.2 Robot horror in popular culture

Robots have long held an ambivalent position in popular culture, oscillating between fascination, utility, and threat. Early works such as Karel Čapek's *R.U.R.* (1920) [59] established robots as figures through which anxieties about autonomy, mechanization, and loss of control could be explored. Contemporary media, including *Ex Machina* (2014) [24] and *M3GAN* (2022) [28], extend this tradition by portraying robots with a near-human form, opaque intentions, and unpredictable actions that generate unease or fear. Analyses of horror narratives show that human-like or semi-human concepts, such as devils or clowns, are frequently used to unsettle audiences, whereas non-human or static concepts, such as cars, are less emotionally evocative [25]. The origins of coulrophobia (fear of clowns), for example, involve facial appearance, unpredictable behavior, and negative media portrayals [51]. These narratives repeatedly mobilize well-documented fear triggers, including ambiguous agency, violations of social norms, bodily vulnerability, and uncanny near-human likeness.

Crucially, robot horror in fiction is not merely a byproduct of the uncanny valley. While uncanny valley research in HRI typically seeks to reduce discomfort, popular culture often amplifies it intentionally. Robot horror often emerges not simply from imperfect human likeness but from the disruption of relational scripts: robots that step too close, mimic emotion without feeling, or behave intelligibly until they suddenly do not. These moments of designed unpredictability create a safe but embodied sense of threat, allowing audiences to experience fear while recognizing the absence of real danger. For HRI and design research, these cultural forms matter because they exemplify fear-based, time-bounded, and recreational interactions that deliberately invite discomfort, providing insight for designing robots capable of evoking high-arousal responses safely.

Immersive storytelling offers a rich source of inspiration for recreational fear-evoking robot design. Experiences that combine technology and human performers, such as *Then She Fell* (Third Rail Projects), *Sleep No More*, *Masquerade*, *Limos* [44], and productions by Darkfield, generate suspense and occasional fear through slow-building narratives, allowing participants to become fully immersed. These experiences often accommodate groups, where shared frightening encounters can amplify engagement and emotional responses, enhancing social bonding [54]. Beyond live performances, immersive technologies can offer additional strategies for eliciting fear. VR haunted houses use haptic stimulation to increase immersion and creepiness [17], while AR games like *Haunted Encounters* at the Natural History Museum of Funchal engage participants in uncovering haunted mysteries through location-based experiences [41]. Collectively, these examples show how popular culture and immersive technology provide design cues and affective strategies that could inform the development of fear-evoking robots.

Situating our work within the lineage of robot horror foregrounds the idea that fear is both culturally constructed and experientially mediated. Robots designed for recreational fear do not interact with neutral users; they interact with audiences primed by decades of narrative and symbolic expectations. Engaging this cultural landscape allows researchers to identify and examine the behavioral components that heighten fear in safe, structured environments.

2.3 Robot horror in theme parks

Theme parks offer a valuable real-world testbed for fear-based robotic experiences, demonstrating both the design strategies and affective mechanisms already used in entertainment settings. Robots have long shaped visitor experiences, from delight to suspense. Early Disney animatronics in attractions like the *Carousel of Progress* [19], along with later examples such as singing birds, dancing pirates, BDX droids [1], and Olaf [2], were designed to entertain and inspire wonder, establishing robots as precise, reliable, and engaging elements of themed environments. Designs before 2000 often featured dinosaurs or monster-like figures, whereas more recent animatronics increasingly portray human-like or child-friendly fictional characters drawn from popular films [31], reflecting both evolving stylistic preferences and technological advances that make humanoid robots more feasible. Forms range from human heads and full-body figures to recognizable characters such as Elsa from Disney's *Frozen*. In HRI research, theme park robots are primarily studied in service roles, such as waiters or attendants [38, 39].

Horror-themed attractions deliberately subvert expectations to elicit fear, with animatronic and robotic figures increasingly incorporated into haunted houses, dark rides, and seasonal scare events. Examples include *Halloween Horror Nights* (e.g., *Chucky 2023*, *Five Nights at Freddy's 2025*) and *Queen Mary's Dark Harbor* (e.g., animatronic patients tied to beds), which employ proximity violations, irregular motion patterns, sudden emotional shifts, and ambiguous responsiveness to create controlled fear experiences. These attractions exemplify visceral theater, leveraging audience instinct, cultural familiarity, and an innate aversion to danger, to provoke strong physiological and emotional responses through deliberate sensory overstimulation without physical contact [9].

These experiences are effective partly because of guests' expectations. Accustomed to smooth, predictable robots in non-horror settings, visitors perceive greater threat when robots move unpredictably, appear too close, or behave unexpectedly.

Across both horror and non-horror contexts, theme parks demonstrate the flexibility of robots as emotional interfaces. Robots can shift between friendly characters, awe-inspiring spectacles, and fear-inducing threats, using mechanical speed, non-biological motion patterns, and the ambiguity of autonomy to shape guest affect. Horror attractions highlight a unique affordance: fear can be elicited in ways distinct from human performers, showing how robotic behavior itself can function as a tool for narrative world-building and emotional modulation. This history of delight and fear provides a fertile context for studying robots designed to evoke unsettling or frightening responses.

2.4 Robot horror in HRI

HRI has not directly investigated robot-induced fear as a design goal, but related research provides insights into creepiness, uncanniness, and emotional discomfort. Studies suggest that components of uncanniness, such as likeability and eeriness, can be shaped by interactions with a robot. Likeability is affected by a robot's attitude towards humans, but repeated interactions can reduce feelings of eeriness regardless of the robot's embodiment [58]. Research with social AI and personal robots further shows that mismatches between user expectations and system behavior can heighten perceptions of creepiness, highlighting the importance of designing interactions that align with users' social norms [27].

Research with children shows that robot morphology plays a key role in perceptions of creepiness. Responses to zoomorphic designs were mixed, with some children finding them familiar and appealing, while others described them as scary or unsettling [18]. Anthropomorphic robots often heightened concerns about emotional volatility or unpredictability, with children fearing that robots could act dangerously or be favored over them by parents. Deviations from expected animal behavior in robot animals, caused by unpredictable or deceptive mimicry, further increased discomfort [56].

To support these efforts, researchers have developed validated scales for assessing perceived creepiness in technology. The seven-item Perceived Creepiness of Voice Assistants Scale (PCAS) measures user discomfort specifically with voice assistants [42], while the Perceived Creepiness of Technology Scale (PCTS) evaluates creepiness across a broader range of devices [55]. PCAS studies suggest that users may abandon voice assistants they perceive as creepy, and other work indicates that a robot may not initially appear unsettling yet still contain unethical design features not fully captured by the PCTS. These findings highlight the importance of incorporating creepiness assessment into HRI design and evaluation to better understand and mitigate user discomfort in social robot interactions [50].

There is also a growing body of HRI research exploring robots for entertainment. Live performance with human actors has been proposed as a testbed for autonomous sociable robots, exemplified by the Public Anemone robot, a highly articulated performer with 13 degrees of freedom that tracked audience movements or

recoiled when approached, demonstrating real-time perception and goal-directed expressive motion [10, 11]. Other projects examined the emotional experience of human participants while combining machine learning, control systems, and artistic strategies to produce compelling robot behaviors. Experiments have examined how changes in movement modes affect participant reactions and gesture choices [20], while swarm intelligence-based robots have been applied in dance performances to achieve precise, coordinated, and smooth collective movements [14]. Together, these studies illustrate how entertainment-focused HRI research can advance both technical capabilities and audience engagement.

3 Discussion

3.1 Fear as an Emotional Dimension in HRI

From an HRI perspective, designing a robot to evoke fear challenges long-standing assumptions about emotional interactions with robots. While some research has examined complex emotions, such as the design and experience of a robot's "death" [29], most HRI research emphasizes minimizing fear and discomfort, prioritizing interpretability, trust, and predictability [4, 22]. When trust or enjoyment decreases and discomfort increases, users may question the robot's competence, potentially reducing acceptance [32]. Consequently, negative emotions such as fear are often treated as undesirable responses rather than informative aspects of interaction, shaping motion, proxemics, appearance, voice, and expressive behavior [5].

Studying fear intentionally, however, can provide insight into how people interpret and evaluate robot behavior. Fear commonly emerges when robots are perceived as threatening [26], ambiguous in intent [3], or in violation of social and spatial expectations [7, 33]. Examining these responses in controlled settings enables researchers to identify perceptual cues such as motion patterns, proxemic intrusion, timing, or responsiveness that trigger threat perceptions. These insights also clarify why everyday robots sometimes provoke unintended anxiety or feelings of creepiness [40], and can inform strategies to reduce such reactions when they are undesirable in practical applications.

We argue that negative emotions, when safely and intentionally elicited, can be meaningful, engaging, and desirable in contexts such as entertainment, art, and therapy [15, 57]. Treating fear as an intentional component of interaction rather than a design failure challenges longstanding assumptions that robots must always be friendly and predictable. This perspective highlights how fear can be used to shape engagement, attention, and experience, and encourages HRI to consider a broader emotional range that includes discomfort as a legitimate and ethically navigable aspect of human-robot interaction.

3.2 Designing Robots to Evoke Fear and High-Arousal Experiences

Designing robots to intentionally evoke fear could refer to insights from domains beyond HRI, such as entertainment, gaming, and psychological research on fear. In theme parks, robots are already used in entertainment settings, including horror-themed attractions, where they typically appear as animatronics with pre-programmed movements, such as in *Five Nights at Freddy's* at Universal Studios

Halloween Horror Nights 2025. These robots often rely on jump scares, loud noises, or sudden movements to elicit fear. Yet, HRI design can move beyond these approaches to explore alternative ways of eliciting fear.

Research from psychology and media studies highlights several physical cues that contribute to fear and tension, as discussed in Section 2.2 on popular culture. In HCI research, horror games are frequently used as experimental stimuli, yet relatively little work has examined what makes a jump scare effective. Evidence suggests that the physiological arousal elicited by a jump scare can be modulated by adjusting the difficulty of concurrent in-game tasks, which require players to engage in substantial mental effort [49]. In a first-person PC horror game, uncanny valley effects and AI behaviors that approach the player from unexpected directions have been shown to heighten fear and tension [36]. These studies are particularly valuable for HRI because they examine fear as a designed outcome, revealing features such as unpredictability, sudden movements, and uncanny appearances or speech patterns that effectively evoke high-arousal responses. While similar behaviors are occasionally observed in HRI research, they typically occur incidentally and are not intended to produce fear or discomfort.

Environmental cues such as spatial design, sound, lighting, and narrative context play a crucial role in creating immersive and frightening experiences. Augmented reality horror games provide useful design insights for high-arousal interaction. For example, users have reported feeling intimidated by virtual robots due to their voices, highlighting how auditory cues can amplify fear [37]. Maintaining thematic consistency between audio and visual elements is also essential, as mismatched sounds, lighting, or visuals can reduce the perceived intensity of the game's atmosphere [45]. Tension arises from situations involving conflict, instability, dissonance, or uncertainty, which trigger predictive processes about what might happen next. When anticipated outcomes differ in their emotional desirability, this divergence generates tension [35]. Researchers designing tension in narrative contexts can also draw on curiosity as a key mechanism shaping suspense. Curiosity functions as both a trigger and an organizing force in how suspense and surprise unfold, and the emotions and cognitive processes generated by this narrative tension are central to the experience of enjoyment [8].

Drawing inspiration from fiction, the movie *Ex Machina* [24] shows how Ava the robot evokes fear through discussions of mortality, desires for freedom, and ambiguous intentions. This suggests that fear can emerge from social and cognitive cues in addition to appearance or sudden movements. Similarly, an autonomous robot could be programmed to engage in unexpected conversations, violate social norms, or behave in ways that suggest ambiguous intent, intentionally creating discomfort or fear. Importantly, such approaches should be carefully calibrated to ensure ethical boundaries and user safety, for instance by limiting the intensity or frequency of norm violations and providing users with clear exit options during interactions.

Practical strategies from live entertainment also offer guidance for safely eliciting fear. Fright attractions such as *Frightmare Farms* in New York use tools, such as the “monster candle,” to signal when to reduce sensory intensity, allowing fear to be modulated to match individual tolerance levels [52]. This system allows fear

to be calibrated to balance safety and enjoyment, an approach that can be applied to interactive robots. For example, a robot could track facial expressions or physiological cues such as heart rate and adapt its behavior to regulate the level of fear it elicits.

Overall, these findings demonstrate that fear responses can be systematically designed for HRI designers seeking to create controlled, high-arousal experiences with interactive robots.

3.3 Implications for Entertainment

Interactive, autonomous animatronic systems are increasingly being used to create engaging, high-arousal experiences for guests. Researchers suggest that robots should recognize visitors and respond appropriately during interactions, adjusting their behavior to engage naturally and dynamically [12]. Compared to human performers, robots offer capabilities that directly map onto established fear triggers, including precisely repeatable timing, non-human movement signatures, unnatural stillness, superhuman reaction times, and programmable unpredictability. These features allow designers to create experiences that are both controlled and emotionally intense.

Autonomous systems also provide practical advantages in entertainment contexts. They deliver consistent, scalable experiences, ensuring that every guest encounters the intended level of intensity, which is difficult to achieve with human actors. Relying solely on human performers introduces safety risks, such as physical altercations, sexual harassment, or accidents during scares [21, 34, 43, 48]. Robots reduce these hazards while addressing labor demands, although this does not imply that they are replacing humans for budgetary reasons only. Instead, robots allow for distinct experiential design possibilities that are difficult or impossible to achieve with human actors.

The deployment of autonomous robots also presents challenges. High upfront costs for development, programming, and maintenance are required, and safety remains critical: robots must avoid harming guests or themselves and may be less adaptable than humans in emergencies. Careful consideration of these limitations is necessary to ensure that high-arousal experiences remain engaging and safe.

4 Future Study Direction and Conclusion

Future work could explore how robots, combined with ambient cues and narrative buildup, manage cognitive load to optimize fear before key moments. By personalizing fear, detecting individual responses and adjusting robot behavior, robots could create safe, engaging, and emotionally calibrated experiences. Studying fear as a designed interaction modality not only advances entertainment applications but also provides insights that help ensure everyday robots interact safely, ethically and inclusively with people, ultimately empowering society through more thoughtful and human-centered robot design.

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